A3-PS003-TC194

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

Chaudhary Tuba personal-
chaudharytuba83@gmail.com
AIORI Hackathon 3
October 14, 2026

Problem Description

Current Approach

Different Traffic Conditions, One Centralised System

Traffic intersections experience different traffic conditions throughout the day.



Location A
Heavy traffic



Location B
Moderate traffic



Location C
Changing Traffic Patterns

The three locations shown are illustrative examples representing different traffic conditions at different intersections.

Central Cloud Server
All raw traffic data is sent to one server for training and prediction.

Main problems

High Bandwidth Usage
Continuous transmission of raw traffic data can consume significant network bandwidth.

Different Traffic Patterns
A shared model may struggle to learn effectively when intersections have different traffic distributions.

Scalability Issues
Adding more intersections increases the data-handling load on the central server.

Network Dependency
Poor connectivity can interrupt or delay traffic data transmission.

Background

How the Current System Works

- Cameras and sensors collect vehicle counts and speeds at intersections.
- Traffic data is sent to a central server for processing and prediction.
- Continuous data transfer increases bandwidth usage and server workload.
- Different traffic patterns make accurate prediction challenging.
- Poor network connectivity can delay data transmission.

Proposed Approach

- **Local Training:** Each intersection trains an AI model using its own traffic data.
- **Model Updates:** Only model updates are sent to the central server.
- **FedAvg Aggregation:** The server combines updates to create a global model.
- **Global Model Sharing:** The updated model is sent back to each intersection for traffic prediction.

Literature & RFC Review

1. McMahan et al. (2017) — Federated Learning

Communication-Efficient Learning of Deep Networks from Decentralized Data

- **Introduced FedAvg for combining locally trained model updates.**
- **Supports our federated training approach**

2. Li et al. (2018) — Traffic Forecasting

Diffusion Convolutional Recurrent Neural Network (DCRNN)

- **Models spatial and temporal traffic patterns.**
- **Provides background for traffic prediction models.**

3. Perifanis et al. (2024) — Federated Traffic Forecasting

Federated Learning for 5G Base Station Traffic Forecasting

- **Studies federated time-series forecasting under non-IID data.**
- **Compares prediction accuracy and communication costs.**

Research Gap

Existing studies explore traffic forecasting and federated learning. **Our project focuses on comparing centralised and FedAvg-based training under IID and non-IID conditions, while measuring prediction accuracy, communication cost, and convergence in one simulator.**

Proposed Solution

We are developing a Federated Learning simulator to predict traffic flow and evaluate prediction accuracy alongside communication cost.

Key Features

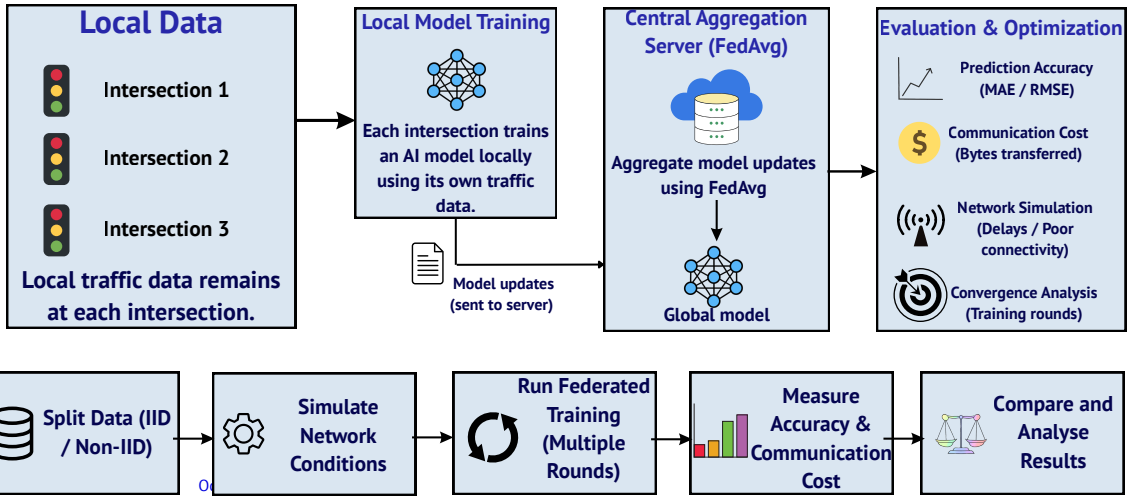
- **Indian Traffic Dataset:** Using vehicle-count data from three signalised intersections in Pune.
- **Traffic Forecasting:** Developing a compact AI model to predict future vehicle counts.
- **IID vs Non-IID Analysis:** Evaluating model performance across similar and different traffic patterns.
- **Convergence Monitoring:** Tracking prediction error across training rounds to measure learning progress.
- **Network Condition Simulation:** Testing the effect of poor connectivity and delayed updates on training.
- **Communication Cost Dashboard:** Visualising data transferred per round alongside prediction accuracy.
- **Comparative Evaluation:** Comparing centralised and federated training using MAE, RMSE, communication cost, and three random seeds.

Optimization Proposed

Current System	Proposed Optimization
Continuous raw data transfer	Exchange model updates instead
Higher potential bandwidth usage	Measure and compare bytes transferred
Centralised training costs	Evaluate federated communication overhead
Network usage not tracked per round	Track cumulative communication cost
Accuracy evaluated separately	Compare prediction accuracy vs bandwidth cost

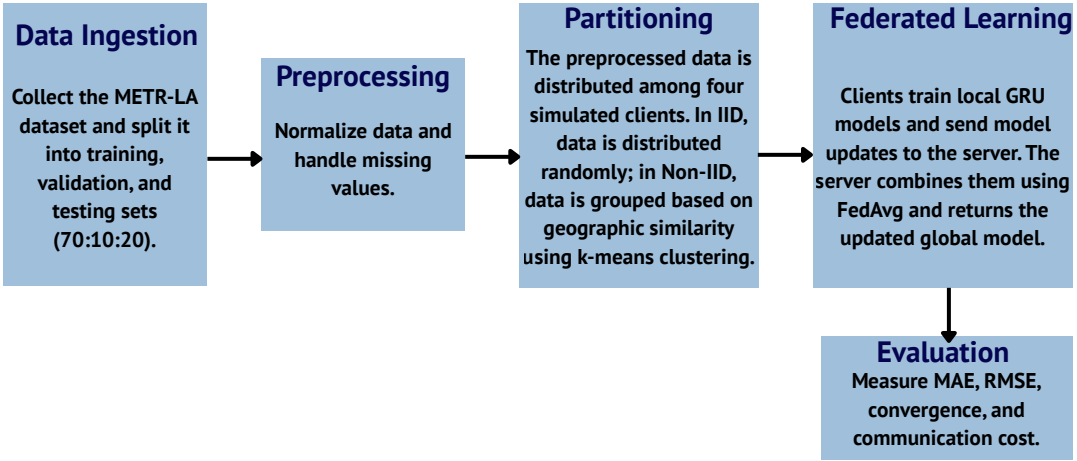
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Data Flow: METR-LA Dataset → Preprocessing → Data Partitioning → Local Training ↔ FedAvg Server → Prediction and Evaluation.



Deployment Timeline

Data Collection



- Collect historical traffic counts from multiple intersections.
- Gather additional data (time, date, etc.).

(Week 1–2)

Data Preprocessing



- Data cleaning.
- Handle missing values.
- Normalization.
- Feature creation (time-based features).

(Week 3)

Data Split — IID / Non-IID



- Split data across 3 clients (intersections).
- Create IID and Non-IID datasets.
- Train-test split locally.

(Week 3)

Model Development



- Implement LSTM model.
- Set up federated learning with FedAvg.
- Configure training parameters.

(Week 4)

Training & Simulation

- Run federated training for multiple rounds.
- Simulate network conditions (delays, packet loss, low bandwidth).
- Track metrics (accuracy, communication cost).

(Week 5)

Evaluation & Deployment



- Predict future vehicle counts.
- Evaluate (MAE / RMSE).
- Compare Federated vs Centralized.
- Analyze results and prepare for deployment.

(Week 6)

Prototype Deployment



- Deploy the trained global model.
- Perform real-time or batch predictions on new traffic data.
- Monitor model performance and communication cost.

(Week 7)

Bibliography I

[1] McMahan, B. et al. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” Proceedings of AISTATS (2017), pp. 1273–1282. https://proceedings.mlr.press/v54/mcmahan17a.html

[2] Li, Y. et al. “Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting.” International Conference on Learning Representations (ICLR) (2018). https://arxiv.org/abs/1707.01926

[3] Perifanis, V. et al. “Federated Learning for 5G Base Station Traffic Forecasting.” Computer Networks (2024). https://doi.org/10.1016/j.comnet.2023.109950

[4] Pune Heterogeneous Traffic Count Dataset. Mendeley Data. https://data.mendeley.com/datasets/xnf2k6n288/1

Thank You for Your Attention!